

Julia Sisk-Reynes

Center for Astrophysics | Harvard & Smithsonian, Cambridge, MA 02138, US
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Research areas: Black holes • Active galactic nuclei • Black hole – galaxy coevolution • Axions • Dark matter
Methods: X-ray spectroscopy • Strong gravitational lensing • Bayesian methods • Machine learning inference

EMPLOYMENT AND RESEARCH

Since Sep 2024 **Postdoctoral Research Fellow**

Center for Astrophysics | Harvard & Smithsonian (CfA), Cambridge, MA
“Using gravitational lenses as high-resolution X-ray telescopes”.
PI: Dr. Daniel Schwartz.

Since Jun 2025 **Co-chair, NewAthena Science Working Group 6: Cosmology & Fundamental Physics**

Chair: Dr. Esra Bulbul (MPE, Garching, Germany).
Co-chair: Dr. Florian Pacaud (University of Bonn, Germany).

EDUCATION

2020 – 2024 **Ph.D. in Astronomy**

Institute of Astronomy (IoA), University of Cambridge, UK
Thesis: “Unveiling fundamental physics with high-resolution X-ray spectroscopy of active galactic nuclei”.
Primary supervisor: Prof. Christopher Reynolds (University of Maryland, College Park).
Secondary supervisor: Dr. James Matthews (University of Oxford, UK).

2016 – 2020 **MPhys in Physics (First Class Honours)**

The University of Manchester, UK
Master’s thesis: “Building models for the thermal Sunyaev-Zel’dovich effect from analytic and numerical cluster pressure profiles”.
Supervisor: Prof. Scott Kay (University of Manchester, UK).

AWARDS AND PRIZES

- 2023 **Two-week Fluid Dynamics Summer School at the Flatiron Institute**
Sponsored by the Center for Computational Astrophysics of the Simons Foundation.
- 2022 **Paul Murdin Prize for the best published paper by a current student**
For [Sisk-Reynes et al., 2022, MNRAS, 510, 1](#). Awarded by the IoA, Cambridge, UK.
- 2020 – 2024 **Full Studentship from the UK Science and Technologies Facilities Council**
To pursue a doctorate in astronomy at the IoA, University of Cambridge.
- Jun 2020 **Stellify Award from The University of Manchester**
Recognition for extracurricular contributions made as an undergraduate student.

SMALL-GROUP TEACHING (TUTORIALS)

Undergraduate supervisions at Cambridge

- 2021– 2022 Mathematics I (Natural Sciences Tripos, first year) – Emmanuel College.
22 weeks, 6 students (2 groups), 1 hour/group/week.
- 2021 Astrophysical Fluid Dynamics (Physics & Astronomy, third year) – Trinity College.
4 weeks, 3 students (1 group), 1.5 hours/week.

Duties: Deliver tutorials, mark weekly problem sets, facilitate discussions, examination preparation.

Peer Assisted Study Support (PASS) Leader, School of Physics and Astronomy at Manchester

- 2017 – 2019 Academic and pastoral weekly support for a total of twenty first-year students.
Duties: Co-lead interactive study support sessions during weekly PASS sessions.

RESEARCH STUDENT SUPERVISION

- Sep 2025 – **Henry Hiscocks**. Southampton – CfA master’s student programme. **Main supervisor**. Incoming Ph.D. student, *Jodrell Bank Centre for Astrophysics, Manchester, UK*. Thesis: “Using MHD simulations to predict photon-axion signals in realistic clusters.” [JSR, Hiscocks et al.](#), in prep. for *NewAthena* Special Issue (JHEAP, 2027).
- Sep 2025 – **Charlotte Eades**. Southampton – CfA master’s student programme. **Co-supervisor**. Incoming Ph.D. student, *Astrophysics Department, University of Oxford, UK*. Thesis: “Unlocking the X-ray structure of the first Einstein zigzag lens J1721”. [Eades, JSR, Schwartz, & Barnacka](#), in prep. (MNRAS, 2026).
- Nov 2024 – **Alex Ostridge**. Full-time exchange master’s student at CfA. **Co-advisor**. Now analyst at Capteiro. Thesis: “Machine learning for milliarcsecond X-ray astrometry in quasar lenses”. [Ostridge, Martinez-Galarza, Schwartz, Sisk-Reynes, & Barnacka](#), in press, MNRAS.
- Jun – Jul 2022 **Esin Gulhabar**. Undergraduate summer research student at IoA, Cambridge. **Advisor**. Former trainee at ESA and now Ph.D. student, *FAU Erlangen-Nurnberg, Germany*. Project: “*XMM* and *NuSTAR* analysis of the broad-line radio galaxy 4C 74.26”.

PRESENTATIONS

34 presentations at international conferences, workshops, telescope collaborations, and group meetings. 17 invited presentations since May 2022. Selected recent invited talks:

- Aug 2026 **Invited conference talk** (*NewAthena* SWG6), COSPAR2026, Florence, Italy
- May 2026 **Invited talk** (lensed quasars), MIT Afternoon Talks, MKI, Cambridge, MA, USA
- Apr 2026 **Invited seminar for a future mission** (massive black hole spins), BHEX AGN SWG
- Apr 2026 **Invited colloquium**, UMBRELA Dialogues, CfA, Cambridge, MA, USA
- Dec 2025 **Invited colloquium**, J. Lin Group Meeting, Hong Kong University
- Oct 2025 **Invited conference talk** (*NewAthena* SWG6), HEAD22/AAS, St Louis, MO, USA
- Oct 2025 **Invited colloquium**, Universidad Nacional Autónoma de México, Mexico
- May 2025 **Invited seminar for a future mission** (*AXIS* spin survey), *AXIS* AGN SWG
- Jan 2025 **Invited colloquium**, AGN group at NASA GSFC, Greenbelt, MD, USA
- Jun 2024 **Invited talk** (astrophysical axion searches), OKC, Stockholm University, Sweden
- Apr 2024 **Invited seminar**, High Energy Astrophysics Group, MPE, Garching, Germany

PRESS COVERAGE

- 2023 *Athena* Press Release ([Sisk-Reynes et al., 2023a](#))
- 2022 *Chandra* Press Release ([Sisk-Reynes et al., 2022b](#))

TELESCOPE TIME

- 2024 **Co-I**, *Chandra* General Observer Program, Cycle 26 – Target: Zw2089 cluster, 640 ks. PI: Dr. Helen Russell (University of Nottingham, UK).
- 2024 **Co-I**, *Chandra* Legacy Program, Cycle 26 – Target: Perseus cluster, 3 Ms. PI: Prof. Andrew Fabian (IoA, University of Cambridge, UK).
- Submitted proposals (awaiting decision):
- Mar 2026 **PI**, *Chandra* Guest Observer Program, Cycle 28 – Target: GrL J0659, 160 ks
- Mar 2026 **PI**, *Chandra* Guest Observer Program, Cycle 28 – Target: DES J2321, 100 ks.
- Mar 2026 **Co-I** (PI D. Schwartz), *Chandra* Guest Observer Program, Cycle 28, 197 ks.
- Jan 2026 **Co-I** (PI D. Schwartz), NASA Astrophysics Data Analysis Program 2026 cycle.
- Jan 2026 **PI**, *NuSTAR* Guest Observer Program, Cycle 12 – Target: 4C 74.26, 150 ks.

SELECTED ACADEMIC SERVICE

Telescope allocation panels

2025 – 2026	<i>Chandra</i> GO Cycles 28 & 29 peer reviewer
2025	<i>XRISM</i> GO Cycle 2 reviewer (3-day in-person panel) – Baltimore, MD, USA
2024	<i>NICER</i> GO Cycle 7 reviewer (5-day remote panel)

Journal paper reviewer

Since 2025	<i>Physical Review D</i> , <i>Physical Review Letters</i> , <i>MNRAS</i>
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Invited panel participation

Jun 2024	Dark matter in Astrophysical Laboratories workshop, <i>KICC</i> , Cambridge, UK
Mar 2024	International Women's Day Panel (for WISE), <i>University of Cambridge</i> , UK

Conference organisation

Jul 2024	Local Organising Committee, IoA 50-year anniversary conference
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SKILLS AND MEMBERSHIP

Programming	Bash, Python, tcl, Unix/Linux
Software	(Py)XSPEC, Sherpa, CIAO, ds9, SAOTrace
Numerical	ALPRO (axion code, advanced level), AREPO (basic level)
Languages	English, Catalan, Spanish (all native/full professional proficiency)
Telescopes	<i>NewAthena</i> (since 2025), <i>AXIS</i> (2022 – 2026), <i>STROBE-X</i> (2022 – 2024)
Membership	IAU (since 2025), AAS/HEAD (since 2024), RAS (2020 – 2024)

SELECTED OUTREACH ACTIVITIES

2025	Guest speaker (Spanish), CfA Public Evening (100 attendees), <i>Cambridge, MA</i>
2025	Guest speaker (Spanish), bilingual school visit (100 attendees), <i>Chelsea, MA</i>
2023 – 2024	Guest speaker , Lindemann Trust Science Day (120 attendees/year), <i>Cambridge, UK</i>

SELECTED INSTITUTIONAL SERVICE

Center for Astrophysics | Harvard & Smithsonian, Cambridge, MA

Since 2025	Lead organiser , Cosmology Journal club
Since 2024	Organiser , High Energy Astrophysics Division seminar series

Institute of Astronomy and University of Cambridge, UK

2024	Exam checker , Part II final examinations (papers 1 – 4)
2021 – 2024	Lead organiser , International Women's Day celebration events
2021 – 2024	MCR seminar speaker series organiser , Newnham College
2021 – 2023	Representative for the public and prospective student applicants
2022 – 2023	Graduate representative , Computer Users Committee
2020 – 2021	Lead organiser , Astronomy Journal Club for first-year graduate physics students
2020 – 2024	Ph.D. cohort representative

University of Manchester, UK

2019 – 2020	Class representative , final-year Physics
2017 – 2020	Regular volunteer , Physics Outreach Society
2017 – 2020	Lead committee member , undergraduate Nuclear & Particle Physics Society Duties: organise monthly seminars and 4-day trip to CERN (50 attendees)

MISSION MEMBERSHIP

Since 2025	<i>NewAthena</i> X-ray Observatory, SWG-6 co-chair and SWG-2 member
2024 – 2026	<i>AXIS</i> , X-ray Phase A probe-class concept proposed to NASA, AGN SWG member
2023 – 2024	<i>STROBE-X</i> , X-ray probe-class concept proposed to NASA, cosmology subgroup member